

常微分

初等积分法

定义

方程

微分方程 $\left\{ \begin{array}{l} \text{常微分: 未知函数一个自变量} \\ \text{偏微分} \end{array} \right.$
 积分...
 线性...: 未知数及其导数次数最多1次
 齐次...: 自变量全移到左右为0

线性=1次 阶=最高阶导数
 齐次=右0 常系数=自变量前

解

①微分方程解—积分曲线

特解—特积分—一条曲线

★不能保证通解, 取任意常数可得特解

通解—通积分—族曲线

任意常数彼此独立 + 其他数 = 阶数 + 完备性

对平面上任一点, 均可确定相应的C, 使该解过该点

②方向场

任意点: 求(x,y)处切线斜率

③定解条件:

一点以函数值+导数值—初始条件—初值问题

不少于两个—边值条件—边值问题

由通解定特解: 条件数 = 任意常数数 = 方程阶数

④解的存在性:

定理1(一阶): 函数及其导数均连续时, 两者相等有唯一解

易错: $\int x dx = \ln|x| + C$

积分原函数在区间而非集合上考虑

故x取值为x<0或x>0而非x≠0

方法

①变量可分离 $\frac{dy}{dx} = f(x)\phi(y)$

$$\phi(y) \neq 0 \Rightarrow \frac{dy}{\phi(y)} = \int f(x) dx + C$$

$$\phi(y) = 0 \Rightarrow \phi(y) = 0 \text{ 以函数 } y=y_0 \text{ 为解}$$

$$\bullet e^x y' = \sqrt{y}(1+y) \Rightarrow e^x \frac{dy}{dx} = \sqrt{y}(1+y)$$

$$\textcircled{1} \sqrt{y}(1+y) \neq 0 \Rightarrow \int \frac{dy}{\sqrt{y}(1+y)} = \int e^x dx \Rightarrow 2 \int \frac{d\sqrt{y}}{1+\sqrt{y}} = \int e^x dx \Rightarrow 2 \arctan \sqrt{y} = e^x + C$$

$$\textcircled{2} \sqrt{y}(1+y) = 0 \Rightarrow y=0 \text{ 或 } y=-1 \text{ 经检验 } y=0 \text{ 为解 默认R内}$$

$$\text{所有解: } 2 \arctan \sqrt{y} = -e^x + C, y=0 (C \text{ 任意})$$

$$\bullet dx + xy dy = y^2 dx + y dy$$

$$(x-1)y dy = (y^2-1) dx$$

$$\textcircled{1} y^2-1 \neq 0 \Rightarrow \int \frac{y}{y^2-1} dy = \int \frac{1}{x-1} dx \Rightarrow \frac{1}{2} \ln|y^2-1| = \ln|x-1| + C \Rightarrow y^2 = C(x-1)^2 + 1 (C = \pm e^{2C} \neq 0)$$

$$\textcircled{2} y^2-1 = 0 \text{ 经检验 } y=\pm 1 \text{ 为解}$$

$$\text{所有解: } y^2 = C(x-1)^2 + 1 (C \text{ 任意})$$

问: 为什么不讨论x=1? → 默认x自变量y的取值

x=1点y=1直线, 点对方程解意义不存

②变量型齐次 $\frac{dy}{dx} = f(\frac{y}{x})$ 令 $u = \frac{y}{x}$

$$\text{令 } u = \frac{y}{x} \Rightarrow \frac{dy}{dx} = \frac{y}{x} \Rightarrow \frac{du}{dx} = \frac{y}{x^2} - \frac{u}{x}$$

$$\bullet \frac{dy}{dx} = \frac{y}{x+y} = \frac{1}{1+\frac{y}{x}} \text{ 令 } u = \frac{y}{x} \Rightarrow \frac{du}{dx} = \frac{1}{1+u} - u$$

$$u \neq 0 \Rightarrow \frac{1}{1+u} - u = \frac{1-u^2}{1+u} \Rightarrow \int \frac{1-u^2}{1+u} du = \int \frac{1}{x} dx$$

$$u=0 \text{ 经检验 } y=0 \text{ 为解}$$

③一阶线性齐次 $\frac{dy}{dx} + p(x)y = 0 \Rightarrow \text{通解 } y = C e^{-\int p(x) dx} (C \text{ 任意})$

$$\bullet \frac{dy}{dx} + p(x)y = 0 \Rightarrow \frac{dy}{dx} = -p(x)y$$

$$y \neq 0 \Rightarrow \frac{1}{y} dy = -p(x) dx \Rightarrow \ln|y| = -\int p(x) dx + C_1 \Rightarrow y = C e^{-\int p(x) dx}$$

$$y=0 \text{ 经检验为解 } \star$$

$$\bullet y' - \frac{1}{x}y = 0 \Rightarrow y = C e^{\int \frac{1}{x} dx} = C|x| = Cx$$

④一阶线性非齐次方程 $\frac{dy}{dx} + p(x)y = f(x) \Rightarrow \text{通解 } y = e^{-\int p(x) dx} \left[\int f(x) e^{\int p(x) dx} dx + C \right] \text{ 3个积分只1个}$

若方程真正解为 $\sin x, u(x) = \sin x e^{\int p(x) dx} \Rightarrow \text{whatever 解, } u(x) \text{ 定}$

$$\text{令 } y = u(x) e^{-\int p(x) dx} \text{ 常数变易法}$$

$$\frac{d}{dx} [u(x) e^{-\int p(x) dx}] + p(x) [u(x) e^{-\int p(x) dx}] = f(x)$$

$$\Rightarrow \frac{du(x)}{dx} e^{-\int p(x) dx} + u(x) e^{-\int p(x) dx} (-p(x)) + p(x) u(x) e^{-\int p(x) dx} = f(x)$$

$$\Rightarrow \frac{du(x)}{dx} e^{-\int p(x) dx} = f(x) \Rightarrow u(x) = \int f(x) e^{\int p(x) dx} dx + C$$

⑤伯努利方程 $\frac{dy}{dx} + p(x)y = f(x)y^n$ $\left\{ \begin{array}{l} n=0 \text{ 一阶线性非齐次} \\ n=1 \text{ 一阶线性齐次} \end{array} \right.$ 以下假设 $n \neq 0, 1$

$$\bullet y^n \frac{dy}{dx} = \frac{1}{n+1} \frac{d}{dx} (y^{n+1}) \text{ 导数降次}$$

$$\textcircled{1} y \neq 0 \Rightarrow y^{-n} \frac{dy}{dx} + p(x)y^{-n+1} = f(x) \Rightarrow \frac{1}{n+1} \frac{d}{dx} (y^{n+1}) + p(x)y^{-n+1} = f(x)$$

$$\text{令 } z = y^{n+1} \Rightarrow \frac{1}{n+1} \frac{dz}{dx} + p(x)z = f(x)$$

$$\textcircled{2} y=0 \star$$

$$\bullet \frac{dy}{dx} = \frac{1}{x} \Rightarrow y = \ln|x| + C$$

辨: $\frac{dx}{dy} = f(x,y)$ 一阶

$\frac{dx}{dy} + p(y) = 0$ 一阶线性

$\frac{dx}{dy} + p(y) = f(y)$ 一阶非线性

$y' = f(\frac{y}{x})$ 变量型齐次

$y' + xy = 0$ 线性齐次

$y' + xy = x$ 线性非齐次

$xy' - y = 0$ 变量型齐次+线性齐次

$$\bullet z' + \frac{1}{z}z = 2$$

$$z = e^{-\int \frac{1}{z} dz} \left[\int 2 e^{\int \frac{1}{z} dz} dz + C \right] = \frac{1}{\ln|z|} (2 \ln|z| + C) = \frac{2}{\ln|z|} + \frac{C}{\ln|z|}$$

$$\bullet (x-y) dx + x dy = 0$$

①一阶线性非齐次公式

②变量型齐次

$$\bullet (x+y) dy = dx$$

$$\textcircled{1} \text{角色互换法 } x = e^{-\int (-1) dy} \left[\int y e^{\int (-1) dy} dy + C \right] = C e^{-y} - y + 1$$

$$\textcircled{2} \text{分离变量法 } \text{令 } u = x+y \Rightarrow u du = dx \Rightarrow u du = (u-x) dx \dots$$

Tip: 验证某函数为方程解

$$dx + xy dy = y^2 dx + y dy$$

$\frac{dy}{dx} + x(y-x) + x^2(y-x)^2 = 1$ 先观察特征 $\Rightarrow \frac{dy-x}{dx} + x(y-x) = -x^2(y-x)^2$

$\frac{dy}{dx} = dx + x^2 dx$
 $\frac{dy}{dx} = x^2 dx + x dx$
 $\Rightarrow y = x$ 不是解

⑥ 全微分方程与积分因子 $Mdx + Ndy = 0 \Rightarrow ydx + xdy = 0 \Rightarrow d(xy) = 0 \Rightarrow xy = C$ 倒推

① 定义: 若 $\exists u(x,y)$ 使 $du = Mdx + Ndy$
 则 $Mdx + Ndy = 0$ 为全微分方程, u 为原函数 $u(x,y) = \int_{(x_0,y_0)}^{(x,y)} Mdx + Ndy$
 使 $Mdx + Ndy$ 有原函数的 M 为积分因子

② 原函数存在条件

$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \Rightarrow M = \frac{\partial u}{\partial x} \quad N = \frac{\partial u}{\partial y} \Rightarrow \frac{\partial M}{\partial y} = \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial N}{\partial x} = \frac{\partial^2 u}{\partial y \partial x}$ 二阶偏导连续时

③ 求原函数

直接凑 $x dx + y dy = 0 \Rightarrow \frac{1}{2} d(x^2 + y^2) = 0 \Rightarrow x^2 + y^2 = C$

积分因子公式 $\frac{dy}{dx} + P(x)y = f(x) \Rightarrow Mdx = e^{\int P(x)dx}$

$du = Mdx + Ndy \Rightarrow \frac{\partial(MN)}{\partial y} = \frac{\partial(NM)}{\partial x} \Rightarrow \frac{\partial M}{\partial y} M + \frac{\partial M}{\partial y} N = \frac{\partial M}{\partial x} N + \frac{\partial N}{\partial x} M$
 $\Rightarrow \frac{\partial M}{\partial y} M - \frac{\partial M}{\partial x} N = M(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y})$

若 M 为 x 函数 $\frac{\partial M}{\partial x} N = M(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}) \Rightarrow \frac{1}{M} \frac{dM}{dx} = \frac{1}{N} (\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x})$
 若 N 为 y 函数 $\frac{\partial M}{\partial y} M = M(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}) \Rightarrow \frac{1}{M} \frac{dM}{dy} = \frac{1}{N} (\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x})$

$x dy - y dx = 0 \Rightarrow y dx - x dy = 0 \Rightarrow M=y \quad N=-x \Rightarrow \frac{1}{N} (\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}) = \frac{1}{-x} (\frac{\partial y}{\partial y} - \frac{\partial (-x)}{\partial x}) = \frac{1}{-x} (1 - (-1)) = \frac{2}{-x}$
 $\Rightarrow \frac{1}{M} \frac{dM}{dx} = \frac{2}{-x} \Rightarrow \ln|M| = -2 \ln|x| \Rightarrow M = \frac{1}{x^2}$

④ 可降阶的二阶微分方程

① $y'' = f(x)$ (直接积分法) 推广到 $y^{(n)} = f(x)$

$y''(t) = g \Rightarrow s = \frac{1}{2} g t^2 + C_1 t + C_2$

② $y'' = f(x,y)$ (不含 y' 型) 令 $p = y' \Rightarrow \frac{dp}{dx} = f(x,p)$

$y'' = 2xy'^2, y'|_{x=0} = -1, y|_{x=0} = 1 \Rightarrow \frac{dp}{dx} = 2xp^2$

$\because p \neq 0 \Rightarrow \int \frac{dp}{p^2} = \int 2x dx \Rightarrow -\frac{1}{p} = x^2 + 1 \Rightarrow p = -\frac{1}{x^2 + 1}$

$\Rightarrow y = -\int \frac{1}{x^2 + 1} dx = -\arctan x + C_2 \Rightarrow y = -\arctan x$

③ $y'' = f(x, y')$ (不含 x 型) 令 $p = y' \Rightarrow y'' = \frac{dp}{dy} \frac{dy}{dx} = p \frac{dp}{dy} \Rightarrow p \frac{dp}{dy} = f(p)$

$y'' = 2y(y')^3, y'|_{x=0} = -1, y|_{x=0} = 0 \Rightarrow \frac{dp}{dy} p = 2yp^3$

$\because p \neq 0 \Rightarrow \int \frac{dp}{p^2} = \int 2y dy \Rightarrow -\frac{1}{p} = y^2 + 1$

$\Rightarrow \frac{dy}{dx} = -\frac{1}{y^2 + 1} \Rightarrow (y^2 + 1) dy = -dx \Rightarrow \frac{1}{3} y^3 + y = -x$

Summary 一阶方程 $\frac{dy}{dx} = f(x,y)$ 30-40%

变量分离 $\frac{dy}{dx} = \varphi(x)\phi(y) \quad e^{xy} y' = \sqrt{y} \ln y$

变量型齐次 $\frac{dy}{dx} = g(\frac{y}{x}) \quad \frac{dy}{dx} = \frac{y}{xy}$

一阶线性 $\frac{dy}{dx} + P(x)y = f(x) \quad y' + \frac{y}{x} = 2 \quad (x+y)dy = dx$

全微分 $Mdx + Ndy = 0 \quad x dx + y dy = 0 \quad x dy - y dx = 0$

算子形式

$$\frac{d}{dx^n} + p_1(x) \frac{d}{dx^{n-1}} + \dots + p_{n-1}(x) \frac{d}{dx} + p_n(x) y = f(x)$$

$f(x) \neq 0$ 非齐次
 $f(x) = 0$ 齐次

齐次通解
非齐次通解

性质: ① 唯一性: 满足初值条件

$$\begin{cases} f(x_0) = y_0^{(0)} \\ f'(x_0) = y_0^{(1)} \\ \vdots \\ f^{(n-1)}(x_0) = y_0^{(n-1)} \end{cases} \Rightarrow \text{唯一解 } y = y_0$$

初值: 1点
边值: ≥2点

线性微分方程

② 线性: $L[\sum_{i=1}^m c_i y_i] = \sum_{i=1}^m c_i L[y_i]$

③ (齐次方程) 叠加原理: 若 $y_1, \dots, y_m$ 为 $L[y] = 0$ 的解, 则 $c_1 y_1 + \dots + c_m y_m$ 也是解

函数组的线性相关: 不全为0的 $a_1, a_2, \dots, a_m$ 使 $a_1 y_1(x) + a_2 y_2(x) + \dots + a_m y_m(x) = 0, x \in (a, b)$ 则线性无关

朗斯基行列式
Wronsky

$$W(x) = \begin{vmatrix} y_1(x) & y_2(x) & \dots & y_m(x) \\ y_1'(x) & y_2'(x) & \dots & y_m'(x) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(m-1)}(x) & y_2^{(m-1)}(x) & \dots & y_m^{(m-1)}(x) \end{vmatrix}$$

解

n 阶齐次线性方程 n 个解 线性相关 $\Leftrightarrow W(x) \equiv 0$

n 个线性无关解为基本解组 $\{y_i(x) | i=1, \dots, n\}$

通解结构 $\begin{cases} n \text{阶齐次线性方程} & Y(x) = \sum_{i=1}^n c_i y_i(x) = Y \\ n \text{阶非齐次线性方程} & Y = Y + Y^* \end{cases}$

二阶常系数

线性微分方程
 $P(x) \equiv P$

预备

① 一元二次方程 $ax^2+bx+c=0$ 的根

② 复数求法 $z^2-2z+6=0$

r 为重根 $ar^2+br+c=0, 2ar+b=0$

$$(z-1)^2+5=0 \Rightarrow z=1 \pm i\sqrt{5}$$

$$z = \frac{2 \pm \sqrt{(-2)^2 - 4 \times 1 \times 6}}{2} = 1 \pm i\sqrt{5}$$

③ 复数表示法

$$z = r(\cos \theta + i \sin \theta) = r \cdot e^{i\theta}$$

④ $y''+py'+qy=0$ 解的性质

若 y_1, y_2 为 $y''+py'+qy=0$ 的解

$$\Rightarrow \begin{cases} y_1''+py_1'+qy_1=0 \\ y_2''+py_2'+qy_2=0 \end{cases} \Rightarrow (y_1+y_2)''+p(y_1+y_2)'+q(y_1+y_2)=0 \Rightarrow y_1+y_2 \text{ 为解}$$

若 y_1 为 $y''+py'+qy=0$ 的解 $\Rightarrow k y_1$ 也为 $y''+py'+qy=0$ 的解

⑤ 如何写出满足指定条件的一个多项式

$$\text{设 } P(x) = 2x^2+3x+10 \text{ 求 } Q(x)$$

$$(1) Q'(x) = P(x) \text{ 设 } Q(x) = ax^3+bx^2+cx = x^2(ax^2+bx+c) \Rightarrow a=8 \ b=3 \ c=5$$

$$(2) Q(x) = P(x) \text{ 设 } Q(x) = ax^2+bx+c = x(ax^2+bx+c) \Rightarrow a=8 \ b=3 \ c=10$$

⑥ $y''+py'+qy=f(x)$ 解的性质 齐+非齐=非齐

若 y_1 为 $y''+py'+qy=0$ 解, $\tilde{y}$ 为 $y''+py'+qy=f(x)$ 解

$$\Rightarrow (y_1+\tilde{y})''+p(y_1+\tilde{y})'+q(y_1+\tilde{y})=f(x) \Rightarrow y_1+\tilde{y} \text{ 为 } y''+py'+qy=f(x) \text{ 解}$$

二阶常系数线性齐次 $y''+py'+qy=0$

猜想解为 $y=e^{rx} \Rightarrow (e^{rx})''+p(e^{rx})'+q e^{rx}=0 \Rightarrow (r^2+pr+q)e^{rx}=0$

$$\Rightarrow r^2+pr+q=0 \text{ 特征方程}$$

① 若 $P^2-4Q>0 \Rightarrow$ 不等实根 $r_1, r_2 \Rightarrow y_1=e^{r_1 x} \ y_2=e^{r_2 x} \Rightarrow y=C_1 e^{r_1 x}+C_2 e^{r_2 x}$

② 若 $P^2-4Q=0 \Rightarrow$ 重实根 $r \Rightarrow y_1=e^{rx}$ 为解 $\Rightarrow$ 设另解为 $y_2=ln(x) e^{rx}$ 带数变易法

$$\text{代入 } [u e^{rx}]''+p[u e^{rx}]'+q[u e^{rx}]=0$$

$$\Rightarrow [u''+(r+p)u'] e^{rx} + u e^{rx}(r^2+pr+q)=0$$

$$\Rightarrow u'' e^{rx}=0 \Rightarrow u''=0 \text{ 取 } u=x$$

$$\Rightarrow y_2=x e^{rx} \Rightarrow y=C_1 y_1+C_2 y_2=C_1 e^{rx}+C_2 x e^{rx}$$

③ 若 $P^2-4Q<0$

$$\text{共轭复根 } \alpha \pm i\beta \text{ 有 } \begin{cases} y_1=e^{\alpha x} \cos \beta x + i e^{\alpha x} \sin \beta x \\ y_2=e^{\alpha x} \cos \beta x - i e^{\alpha x} \sin \beta x \end{cases} \text{ 为解} \Rightarrow \text{通解为 } C_1 e^{\alpha x} \cos \beta x + C_2 e^{\alpha x} \sin \beta x$$

$$y''+4y'-5y=0 \Rightarrow r^2+4r-5=0 \Rightarrow \text{通解 } y=C_1 e^{-5x}+C_2 e^x$$

$$y''+4y'+4y=0 \Rightarrow \text{重根 } -2 \Rightarrow \text{通解 } y=C_1 e^{-2x}+C_2 x e^{-2x}$$

$$y''+4y'-5y=0 \Rightarrow \text{复根 } -2 \pm i\sqrt{3} \Rightarrow \text{通解 } y=e^{-2x}(C_1 \cos \sqrt{3}x + C_2 \sin \sqrt{3}x)$$

二阶常系数线性非齐次 $y''+py'+qy=f(x)$

$$y''+py'+qy=P_m(x)e^{\alpha x} \xrightarrow{\text{m次多项式}}$$

① 设特解为 $\tilde{y} = Q_m(x)e^{\alpha x}$

$$\Rightarrow \tilde{y}''+p\tilde{y}'+q\tilde{y} = (Q_m e^{\alpha x})''+p(Q_m e^{\alpha x})'+q(Q_m e^{\alpha x})$$

② $r^2+pr+q=0$ 特征方程根

非根
 α 为单根
重根

$$y''+py'+qy=P_m(x)e^{\alpha x} \text{ 的特解 } y=Q_m(x)e^{\alpha x}$$

$Q=R_m(x)$
 $Q=XR_m(x)$
 $Q=X^2R_m(x)$

线性微分方程

$$= (R'' + 2\alpha R' + \alpha^2 R)e^{ax} + p(R' + \alpha R)e^{ax} + qRe^{ax}$$

$$= [R'' + (2\alpha + p)R' + (\alpha^2 + p\alpha + q)R]e^{ax} = P_m(x)e^{ax}$$

$$\Rightarrow R'' + (2\alpha + p)R' + (\alpha^2 + p\alpha + q)R = P_m(x)$$

$$\Rightarrow R = R_m(x)$$

$$y'' + py' + qy = P_m(x)e^{ax} \text{ 的通解}$$

$$\Leftrightarrow y'' + py' + qy = 0 \text{ 通解} + y'' + py' + qy = P_m(x)e^{ax} \text{ 的特解} \hat{y} = R_m(x)e^{ax}$$

- ① 若 $\alpha^2 + p\alpha + q \neq 0 \Rightarrow R = R_m(x)$
- ② 若 $\alpha^2 + p\alpha + q = 0$ 且 $2\alpha + p \neq 0 \Rightarrow R = xR_m(x)$
- ③ 若 $\alpha^2 + p\alpha + q = 0$ 且 $2\alpha + p = 0 \Rightarrow R = x^2R_m(x)$

● $y'' + 4y' - 5y = 128e^{2x}$ 通解

$\alpha = 2$ 不为 $r^2 + 4r - 5 = 0$ 根 $\Rightarrow$ 设特解 $\hat{y} = (ax+b)e^{2x}$

$\Rightarrow 16(ax+b) + 40a = 128x \Rightarrow a=8, b=-5 \Rightarrow \hat{y} = (8x-5)e^{2x}$

$\therefore$ 通解 $y = (8x-5)e^{2x} + C_1e^{-x} + C_2e^x$

● $y'' + 4y' - 5y = 36xe^x$

$\alpha = 1$ 为单根 $\Rightarrow$ 特解 $\hat{y} = x(ax+b)e^x$

自由项 $f(x) = P_m(x)e^{ax}\cos bx$ $f(x) = P_m(x)e^{ax}\sin bx$

● $\begin{cases} y_1'' + py_1' + qy_1 = P_m(x)e^{ax}\cos bx \\ y_2'' + py_2' + qy_2 = P_m(x)e^{ax}\sin bx \end{cases} \Rightarrow \begin{cases} y_1'' + py_1' + qy_1 = P_m(x)e^{ax}\cos bx \\ (y_2'' + py_2' + qy_2)i = [P_m(x)e^{ax}\sin bx]i \end{cases}$ 转化为已知类型

$\Rightarrow y'' + py' + qy = P_m(x)e^{ax}(\cos bx + i\sin bx) \Rightarrow y'' + py' + qy = P_m(x)e^{(a+bi)x} \Rightarrow y'' + py' + qy = P_m(x)e^{ax} \quad (\alpha = a+bi)$

● $y'' - y = e^x \sin 2x$

① 特解 $\begin{cases} y'' - y = e^x \cos 2x \\ y'' - y = e^x \sin 2x \end{cases} \Rightarrow y'' - y = e^{ix} \text{ 将取解 } y \text{ 的虚部}$

② 求 $y'' - y = 0$ 通解 $r^2 - 1 = 0$ 通解 $= C_1e^x + C_2e^{-x}$

③ 求 $y'' - y = e^{ix}$ 特解 设 $\hat{y} = ke^{ix} \Rightarrow (ke^{ix})'' - ke^{ix} = e^{ix} \Rightarrow (k\alpha^2 - k)e^{ix} = e^{ix} \Rightarrow k = \frac{1}{\alpha^2 - 1}$

$\therefore$ 特解 $\hat{y} = \frac{1}{-4+1}e^{(1+2i)x} = -\frac{1}{3}e^x(\cos 2x + i\sin 2x) = -\frac{1}{3}e^x[\cos 2x - \sin 2x] + i[\cos 2x + \sin 2x]$

结论: 通解 $= C_1e^x + C_2e^{-x} - \frac{1}{3}e^x(\cos 2x + \sin 2x)$

① 欧拉方程: $x^n y^{(n)} + a_1 x^{n-1} y^{(n-1)} + \dots + a_{n-1} x y' + a_n y = f(x)$ $f(x)$ 连续

● $x^2 y'' + x y' + 4y = 8x^2 \quad (x > 0)$

① 换元 $x = e^t \Rightarrow t = \ln x$

$y' = \frac{dy}{dx} = \frac{1}{x} \frac{dy}{dt} \Rightarrow xy' = \frac{dy}{dt}$

$y'' = \frac{d^2y}{dx^2} = \frac{d}{dx}(\frac{1}{x} \frac{dy}{dt}) = -\frac{1}{x^2} \frac{dy}{dt} + \frac{1}{x^2} \frac{d^2y}{dt^2} \Rightarrow x^2 y'' = -\frac{dy}{dt} + \frac{d^2y}{dt^2}$

② 代入 $\Rightarrow \frac{d^2y}{dt^2} + 4y = 8e^{2t}$

通解 $= y = C_1 \cos 2t + C_2 \sin 2t + e^{2t}$

Tip 算子法 $D = \frac{d}{dt}$

$\begin{cases} xy' = Dy \\ x^2 y'' = D(D-1)y \\ x^3 y''' = D(D-1)(D-2)y \end{cases} \Rightarrow x^n y^{(n)} = D(D-1)\dots(D-n+1)y$

代入 $D(D-1)y + Dy + 4y = 8e^{2t}$

$\Rightarrow D^2 y + 4y = 8e^{2t}$

变系数
变量代换

② 常数变易法

I. 齐次 $y'' + p(x)y' + q(x)y = 0 \Rightarrow$ 通解 $= y_1 [C_1 \int |y_1|^{-2} e^{-\int p(x) dx} dx + C_2]$ 与 $q(x)$ 无关!

● 特解 y_1 易知 $y = C y_1$ 为解 常数变易法 $\boxed{y = u y_1}$

$\Rightarrow (u'' y_1 + 2u' y_1' + u y_1'') + p(x)(u' y_1 + u y_1') + q(x) u y_1 = 0$

$\Rightarrow [u'' y_1 + 2u' y_1' + p(x) u y_1'] + u [y_1'' + p(x) y_1' + q(x) y_1] = 0$

$\Rightarrow u'' y_1 + 2u' y_1' + p(x) u y_1' = 0$ 变量分离 $\Rightarrow u'' + [2\frac{y_1'}{y_1} + p(x)] u' = 0$

$\boxed{z = u'} \Rightarrow z' + [2\frac{y_1'}{y_1} + p(x)] z = 0$ Tip $y' + \square y = 0 \Rightarrow (e^{\int \square dx} \cdot y)' = 0$

$\Rightarrow z = C_1 e^{-\int [2\frac{y_1'}{y_1} + p(x)] dx} = C_1 |y_1|^{-2} e^{-\int p(x) dx} \Rightarrow u = C_1 \int |y_1|^{-2} e^{-\int p(x) dx} dx + C_2$

II. 非齐次 $y'' + p(x)y' + q(x)y = f(x)$

已知齐次通解 $C_1 y_1 + C_2 y_2 \Rightarrow$ 求真特解 $\hat{y} = u_1 y_1 + u_2 y_2 \begin{cases} u_1' y_1 + u_2' y_2 = 0 \\ u_1' y_1' + u_2' y_2' = f(x) \end{cases}$

● 没特解为 $\hat{y} = u_1 y_1 + u_2 y_2$ 加限定

$\hat{y}' = (u_1' y_1 + u_1 y_1') + (u_2' y_2 + u_2 y_2') = (u_1' y_1 + u_2' y_2) + (u_1 y_1' + u_2 y_2')$

● $y'' + \frac{1}{x} y' - \frac{1}{x^2} y = 0$

① 猜个特解 $\Rightarrow x^2 y'' + x y' - y = 0$ 猜 x^λ

$\Rightarrow x^2 \lambda(\lambda-1)x^{\lambda-2} + x \lambda x^{\lambda-1} - x^\lambda = 0$

$\Rightarrow [\lambda(\lambda-1) + \lambda - 1] x^\lambda = 0 \Rightarrow \lambda = \pm 1$ 取 $y = x$

② 再用 Liouville 求通解 $p(x) = \frac{1}{x}$

$y = x [C_1 \int \frac{1}{x^2} e^{-\int \frac{1}{x} dx} dx + C_2]$

$= x [C_1 \int \frac{1}{x^2} dx + C_2]$

$= \begin{cases} x (\frac{1}{2} C_1 x^2 + C_2) & x > 0 \\ x (-\frac{1}{2} C_1 x^2 + C_2) & x < 0 \end{cases} = C_1 x^3 + C_2 x$

● $y'' - 3y' + 2y = \frac{1}{1-e^x}$ 通解

① 先求 $y'' - 3y' + 2y = 0$ 通解

$y = C_1 e^x + C_2 e^{2x}$

令 $u_1' y_1 + u_2' y_2 = 0 \Rightarrow \tilde{y}' = u_1 y_1' + u_2 y_2' \Rightarrow \tilde{y}'' = (u_1' y_1'' + u_1 y_1''') + (u_2' y_2'' + u_2 y_2''')$

代入 $\Rightarrow [(u_1' y_1'' + u_1 y_1''') + (u_2' y_2'' + u_2 y_2''')] + p_{21}(u_1 y_1' + u_2 y_2') + q_{21}(u_1 y_1 + u_2 y_2) = f(x)$
 $\Rightarrow [u_1 y_1'' + p_{21} u_1 y_1' + q_{21} u_1 y_1] + [u_2 y_2'' + p_{21} u_2 y_2' + q_{21} u_2 y_2] + (u_1' y_1'' + u_2' y_2'') = f(x)$
 $\Rightarrow u_1' y_1'' + u_2' y_2'' = f(x)$

② 记特解为 $y = u_1 e^x + u_2 e^{2x}$

$$\begin{cases} u_1' e^x + u_2' e^{2x} = 0 \\ u_1' e^x + 2u_2' e^{2x} = 0 \end{cases} \Rightarrow \begin{cases} u_1' = -\frac{e^x}{1+e^x} \\ u_2' = \frac{e^{-x}}{1+e^x} \end{cases} \Rightarrow \begin{cases} u_1 = \ln(1+e^x) \\ u_2 = -e^{-x} + \ln(1+e^x) \end{cases}$$

③ 原方程通解
 $y = C_1 e^x + C_2 e^{2x} + (e^x + e^{2x}) \ln(1+e^x)$

Summary = 阶方程 $y'' = f(x, y, y')$ 30-40%

可降阶 $y'' = \cos x$ $y'' = 2x(y')^2$ $y'' = 2y(y')^3$

= 阶常系数线性 $y'' + 4y' - 5y = 0$ $y'' + 4y' - 5y = 16e^{2x}$

$y'' + 4y' + 4y = 0$ $y'' + 4y' + 4y = 6xe^{2x}$

$y'' + 4y' + 7y = 0$ $y'' - y = e^x \sin 2x$

= 阶变系数线性 $x^2 y'' + x y' + 4y = x^2$ Euler

$y'' + \frac{1}{x} y' - \frac{1}{x^2} y = 0$ Liouville

$y'' - 3y' + 2y = \frac{1}{1+e^x}$ 常数变易

变形及向高阶的推广 $P_{34} \rightarrow P_{30} - 15$ $P_{20} - 121$

补: 1重特征根

① A 的 1 重 $\lambda = 0$ $\vec{v} e^{0x}$ 为 1 重解

② A 的 1 重 λ_0 解: $\vec{x} = (\vec{u} + t\vec{v}) e^{\lambda_0 t}$ 其中 $\vec{u} = (A - \lambda_0 E)\vec{v}$

$\vec{u}, \vec{v}$ 为满足 $(A - \lambda_0 E)^2 \vec{u} = 0$ 的 1 重-向量

③ ... = ... $\vec{x} = (\vec{u} + \frac{1}{1!} t\vec{v}_1 + \frac{1}{2!} t^2 \vec{v}_2) e^{\lambda_0 t}$

$\vec{u} = (A - \lambda_0 E)\vec{v}$ $\vec{v}_2 = (A - \lambda_0 E)^2 \vec{v}$

$\vec{u}, \vec{v}$ 为满足 $(A - \lambda_0 E)^2 \vec{u} = 0$ 的 1 重-向量

④ $A = \begin{bmatrix} 0 & 1 \\ -9 & 6 \end{bmatrix} \Rightarrow \lambda = 3$

① $(A - \lambda_0 E)^2 \vec{v} = 0$ 取线性无关 $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\vec{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$\vec{v}_1 \rightarrow \vec{x}_1 = (\vec{u} + t\vec{v}_1) e^{\lambda_0 t}$

$\vec{v}_2 \rightarrow \vec{x}_2 = (\vec{u} + t\vec{v}_2) e^{\lambda_0 t}$

通: $C_1 \vec{x}_1 + C_2 \vec{x}_2$

② 消元 $\Rightarrow x'' - 6x' + 9x = 0$

补: 幂级数法

④ $\frac{dy}{dx} = y$ 令 $y = a_0 + a_1 x + \dots$

$\Rightarrow a_0 + a_1 x + a_2 x^2 + \dots = a_1 + 2a_2 x + 3a_3 x^2 + \dots$

$\begin{cases} a_0 = a_1 \\ a_1 = 2a_2 \\ \vdots \end{cases} \Rightarrow \begin{cases} a_1 = a_0 \\ a_2 = a_0/2 \\ \vdots \end{cases} y = a_0 (1 + \frac{1}{1!} x + \frac{1}{2!} x^2 + \dots) \Rightarrow y = C e^x$

n元微分方程组 $\Leftrightarrow$ n阶微分方程

④ $x''' - 3x'' + 2x' + x = \sin t$ $\begin{cases} \frac{dx_1}{dt} = x_2 \\ \frac{dx_2}{dt} = x_3 \\ \frac{dx_3}{dt} = 3x_3 - 2x_2 - x_1 + \sin t \end{cases}$

线性微分方程组

$\begin{cases} \frac{dx_1}{dt} = \sum_{j=1}^n a_{1j}(t)x_j + f_1(t) \\ \vdots \\ \frac{dx_n}{dt} = \sum_{j=1}^n a_{nj}(t)x_j + f_n(t) \end{cases} \Leftrightarrow \frac{d\vec{x}}{dt} = A(t)\vec{x} + \vec{f}(t)$ $A(t) = \begin{bmatrix} a_{11}(t) & a_{12}(t) & \dots & a_{1n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}(t) & a_{n2}(t) & \dots & a_{nn}(t) \end{bmatrix}$ $\vec{x} = \begin{bmatrix} x_{1(t)} \\ \vdots \\ x_{n(t)} \end{bmatrix}$ $\vec{f}(t) = \begin{bmatrix} f_{1(t)} \\ \vdots \\ f_{n(t)} \end{bmatrix}$

唯一: $\frac{d\vec{x}}{dt} = A(t)\vec{x} + \vec{f}(t)$ $A(t), \vec{f}(t)$ 在 (a,b) 内连续, 则 初值问题 $\vec{x}(t_0) = \vec{x}_0$ $t_0 \in (a,b)$ 内存在唯一-零解

组合: $C_1 \vec{x}_1(t) + C_2 \vec{x}_2(t) + \dots + C_n \vec{x}_n(t) = 0$

④ 证 $\vec{u}_1 e^{\lambda_1 t}, \vec{u}_2 e^{\lambda_2 t}, \dots, \vec{u}_n e^{\lambda_n t}$ 在 (a, ∞) 上线性无关, 其中 $\lambda_1, \dots, \lambda_n$ 互不相同, $\vec{u}_1, \dots, \vec{u}_n$ 全非 0

Wronsky 行列式 $[\vec{x}_1(t), \vec{x}_2(t), \dots, \vec{x}_n(t)] = 0$ $t \in (a,b)$

线性相关 $\Leftrightarrow [\vec{x}_1(t), \dots, \vec{x}_n(t)] = 0$ 无关 $\Leftrightarrow \neq 0$

$\vec{x}_1(t), \dots, \vec{x}_n(t)$ 为解 $\Leftrightarrow [\dots]$ 恒为 0 / 非 0

解 = A(t) x 通解结构

线性无关解 $\vec{x}_1(t), \dots, \vec{x}_n(t)$ 基本解组 $[\dots]$ 基本解矩阵

存在: 必有且最多只有 n 个线性无关解

① 二阶常系数线性方程

$$\begin{cases} \frac{dx}{dt} = 5x - 2y + e^{2t} \\ \frac{dy}{dt} = 6x - 2y \end{cases} \xrightarrow{\text{消元}} \begin{cases} 2y = 5x - \frac{dx}{dt} + e^{2t} \\ \frac{d}{dt}(2y) = 12x - 2\frac{dx}{dt} \end{cases}$$

$$\Rightarrow \frac{d}{dt}(5x - \frac{dx}{dt} + e^{2t}) = 12x - 2\frac{dx}{dt} \Rightarrow \frac{dx}{dt} - 3\frac{dx}{dt} + 2x = 5e^{2t}$$

$$\Rightarrow x = C_1 e^t + C_2 e^{2t} + \frac{5}{2} e^{2t} \Rightarrow y = 2C_1 e^t + \frac{3}{2} C_2 e^{2t} + e^{2t} \quad \text{回到} y!!$$

② 二阶常系数线性齐次方程组的特征向量方法

$$\begin{cases} \frac{dx}{dt} = a_{11}x + a_{12}y \\ \frac{dy}{dt} = a_{21}x + a_{22}y \end{cases} \quad \text{猜想} \quad \begin{cases} x = k_1 e^{\lambda t} \\ y = k_2 e^{\lambda t} \end{cases} \Rightarrow \begin{cases} k_1 \lambda = a_{11}k_1 + a_{12}k_2 \\ k_2 \lambda = a_{21}k_1 + a_{22}k_2 \end{cases}$$

$$\Rightarrow \lambda \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} \quad \lambda \text{ 为 } \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \text{ 的特征值}$$

$$\begin{bmatrix} k_1 \\ k_2 \end{bmatrix} \text{ 为特征向量 } (\lambda E - A) \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = 0$$

$$\begin{cases} \frac{dx}{dt} = 5x - 2y \\ \frac{dy}{dt} = 6x - 2y \end{cases}$$

① 写矩阵, λ $A = \begin{bmatrix} 5 & -2 \\ 6 & -2 \end{bmatrix}$ $\lambda = 1$ 或 $2 \Rightarrow \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

② 写特解 $\begin{bmatrix} 1 \\ 2 \end{bmatrix} e^t \begin{bmatrix} 2 \\ 3 \end{bmatrix} e^{2t}$

③ 写通解 $\begin{bmatrix} x \\ y \end{bmatrix} = C_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^t + C_2 \begin{bmatrix} 2 \\ 3 \end{bmatrix} e^{2t} = \begin{bmatrix} C_1 e^t + 2C_2 e^{2t} \\ 2C_1 e^t + 3C_2 e^{2t} \end{bmatrix}$

求A的所有特征值 λ_1, λ_2 及对应的线性无关特征向量 $\vec{v}_1, \vec{v}_2$

特解为 $\vec{x}_1 = \vec{v}_1 e^{\lambda_1 t}$ $\vec{x}_2 = \vec{v}_2 e^{\lambda_2 t}$

通解为 $\vec{x} = C_1 \vec{x}_1 + C_2 \vec{x}_2 = C_1 \vec{v}_1 e^{\lambda_1 t} + C_2 \vec{v}_2 e^{\lambda_2 t}$ 可推广到三元以上 $\Rightarrow$ 通解: $(x_1, \dots, x_n) \begin{pmatrix} C_1 \\ \vdots \\ C_n \end{pmatrix}$

补充: 实矩阵的复特征根与特征向量 $\rightarrow$ 都取共轭

$$A\vec{z} = \lambda\vec{z} \Rightarrow A\bar{\vec{z}} = \bar{\lambda}\bar{\vec{z}} \quad (\text{取共轭}) \Rightarrow A\bar{\vec{z}} = \bar{\lambda}\bar{\vec{z}}$$

$\Rightarrow \bar{\vec{z}}$ 为A的属于 $\bar{\lambda}$ 的特征向量

$$\begin{cases} \frac{dx}{dt} = -y \\ \frac{dy}{dt} = x \end{cases} \quad \text{特征向量法}$$

① 写矩阵 $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ $\lambda = i, -i$ 对应的 $\vec{z}$ 为 $\begin{bmatrix} i \\ 1 \end{bmatrix} \begin{bmatrix} -i \\ 1 \end{bmatrix}$

② 写特解 $\vec{x}_1 = \begin{bmatrix} i \\ 1 \end{bmatrix} e^{it}$ $\vec{x}_2 = \begin{bmatrix} -i \\ 1 \end{bmatrix} e^{-it}$

③ 实虚部分离 $\vec{x}_1 = \begin{bmatrix} i \\ 1 \end{bmatrix} e^{it} = \begin{bmatrix} i \\ 1 \end{bmatrix} (\cos t + i \sin t) = \begin{bmatrix} -\sin t \\ \cos t \end{bmatrix} + i \begin{bmatrix} \cos t \\ \sin t \end{bmatrix}$

$$\vec{x}_2 = \begin{bmatrix} -i \\ 1 \end{bmatrix} e^{-it} = \begin{bmatrix} -i \\ 1 \end{bmatrix} (\cos t - i \sin t) = \begin{bmatrix} \sin t \\ \cos t \end{bmatrix} - i \begin{bmatrix} \cos t \\ \sin t \end{bmatrix}$$

④ 写实值函数特解 $\frac{1}{2}(\vec{x}_1 + \vec{x}_2) = \begin{bmatrix} -\sin t \\ \cos t \end{bmatrix}$ $\frac{1}{2i}(\vec{x}_1 - \vec{x}_2) = \begin{bmatrix} \cos t \\ \sin t \end{bmatrix}$ 这种对应是必然的 可直接写出

⑤ 写通解 $\begin{bmatrix} x \\ y \end{bmatrix} = C_1 \begin{bmatrix} -\sin t \\ \cos t \end{bmatrix} + C_2 \begin{bmatrix} \cos t \\ \sin t \end{bmatrix} = \begin{bmatrix} -C_1 \sin t + C_2 \cos t \\ C_1 \cos t + C_2 \sin t \end{bmatrix}$

转为二阶方程

$$\xrightarrow{\text{消元}} \frac{d}{dt} \left[\frac{dx}{dt} \right] = x \Rightarrow x'' + x = 0$$

$$\Rightarrow x = C_1 \sin t + C_2 \cos t$$

$$\Rightarrow y = -x' = -C_1 \cos t + C_2 \sin t$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} C_1 \sin t + C_2 \cos t \\ -C_1 \cos t + C_2 \sin t \end{bmatrix}$$

③ 一阶常系数线性非齐次方程组的常数变易法

补充: 记号与运算

矩阵的导数 $[a_{ij}(t)]' = [a'_{ij}(t)]$

向量的导数 $\begin{bmatrix} x \\ y \end{bmatrix}' = \vec{x}'$ 且 $\vec{x}' \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x' \\ y' \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$

若 $\vec{x}_1, \vec{x}_2$ 为 $\frac{d}{dt} \vec{x} = A\vec{x}$ 的解 $\Rightarrow \frac{d}{dt} \vec{x}_1 = A\vec{x}_1$ $\frac{d}{dt} \vec{x}_2 = A\vec{x}_2$

记 $[\vec{x}_1, \vec{x}_2] = X$ 为基本解矩阵, $\begin{bmatrix} C_1(t) \\ C_2(t) \end{bmatrix} = \vec{C}(t)$

$$X' = \frac{d}{dt} [\vec{x}_1, \vec{x}_2] = [A\vec{x}_1, A\vec{x}_2] = AX$$

$$\frac{d}{dt} [X\vec{C}(t)] = \frac{d}{dt} \left\{ \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} C_1(t) \\ C_2(t) \end{bmatrix} \right\} = \frac{d}{dt} [C_1(t)x_1 + C_2(t)x_2] = [C_1'(t)x_1 + C_2'(t)x_2] + [C_1(t)x_1' + C_2(t)x_2'] = [\vec{x}_1, \vec{x}_2] \begin{bmatrix} C_1'(t) \\ C_2'(t) \end{bmatrix} + [X\vec{C}(t)]' = [X\vec{C}(t)]' + [X\vec{C}(t)]' = X\vec{C}(t) + X'\vec{C}(t) = \text{函数之积的导数的推广}$$

可推广到三元以上

求非齐次的特解

$$\begin{cases} \frac{dx}{dt} = a_{11}x + a_{12}y + f_1(t) \\ \frac{dy}{dt} = a_{21}x + a_{22}y + f_2(t) \end{cases} \quad \text{记 } \frac{d}{dt} \vec{x} = A\vec{x} + \vec{f}(t) \quad \left(\text{其中 } A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \vec{x} = \begin{bmatrix} x \\ y \end{bmatrix} \vec{f}(t) = \begin{bmatrix} f_1(t) \\ f_2(t) \end{bmatrix} \right)$$

若齐次通解为 $\vec{x} = C_1 \vec{x}_1 + C_2 \vec{x}_2$

记非齐次特解为 $\vec{x}^* = C_1(t) \vec{x}_1 + C_2(t) \vec{x}_2 = [\vec{x}_1, \vec{x}_2] \begin{bmatrix} C_1(t) \\ C_2(t) \end{bmatrix} = X\vec{C}(t)$

$$\Rightarrow \frac{d}{dt} (X\vec{C}(t)) = X\vec{C}'(t) + X'\vec{C}(t) = AX\vec{C}(t) + \vec{f}(t)$$

$$\Rightarrow X\vec{C}'(t) = \vec{f}(t) \Rightarrow \vec{C}'(t) = X^{-1}\vec{f}(t)$$

写出通解

$$\frac{d}{dt} \vec{x} = A\vec{x} \quad \text{通解: } \vec{x} = C_1 \vec{x}_1 + C_2 \vec{x}_2$$

$$\frac{d}{dt} \vec{x} = A\vec{x} + \vec{f}(t) \quad \text{特解: } \vec{x}^* = C_1(t) \vec{x}_1 + C_2(t) \vec{x}_2$$

$$\frac{d}{dt} \vec{x} = A\vec{x} + \vec{f}(t) \quad \text{通解: } \vec{x} = C_1 \vec{x}_1 + C_2 \vec{x}_2 + \vec{x}^*$$

$$\begin{cases} \frac{dx}{dt} = 5x - 2y + e^{2t} \\ \frac{dy}{dt} = 6x - 2y \end{cases} \quad \text{常数变易法}$$

① 写矩阵 $\frac{d}{dt} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 & -2 \\ 6 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} e^{2t} \\ 0 \end{bmatrix}$

齐次通解 $\begin{bmatrix} x \\ y \end{bmatrix} = C_1 \vec{x}_1 + C_2 \vec{x}_2 \quad X = [\vec{x}_1, \vec{x}_2] = \begin{bmatrix} e^t & 2e^{2t} \\ 2e^t & 3e^{2t} \end{bmatrix}$

$$X \begin{bmatrix} C_1' \\ C_2' \end{bmatrix} = \vec{f}(t) \Rightarrow \begin{bmatrix} e^t & 2e^{2t} \\ 2e^t & 3e^{2t} \end{bmatrix} \begin{bmatrix} C_1' \\ C_2' \end{bmatrix} = \begin{bmatrix} e^{2t} \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} C_1' \\ C_2' \end{bmatrix} = \begin{bmatrix} e^{2t} \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} C_1' \\ C_2' \end{bmatrix} = \begin{bmatrix} -e^{2t} \\ e^{2t} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}e^{2t} \\ e^{2t} \end{bmatrix} \Rightarrow \begin{bmatrix} C_1(t) \\ C_2(t) \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}e^{2t} \\ e^{2t} \end{bmatrix}$$

消元法

$$\xrightarrow{\text{消元}} \begin{cases} 2y = 5x - \frac{dx}{dt} + e^{2t} \\ \frac{d}{dt}(2y) = 12x - 2\frac{dx}{dt} \end{cases} \Rightarrow \frac{dx}{dt} - 3\frac{dx}{dt} + 2x = 5e^{2t}$$

$$\Rightarrow x = C_1 e^t + C_2 e^{2t} + \frac{5}{2} e^{2t}$$

$$\Rightarrow y = 2C_1 e^t + \frac{3}{2} C_2 e^{2t} + e^{2t}$$

非齐次特解 $\vec{x} = C_1(t)\vec{r}_1 + C_2(t)\vec{r}_2 = -13e^{2t} \begin{bmatrix} e^{2t} \\ 2e^{2t} \end{bmatrix} + 2e^t \begin{bmatrix} 2e^{2t} \\ 3e^{2t} \end{bmatrix} = \begin{bmatrix} 23e^{2t} \\ 3e^{2t} \end{bmatrix}$

齐次通解 $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} C_1 e^t + 2C_2 e^{2t} \\ 2C_1 e^t + 3C_2 e^{2t} \end{bmatrix}$

非齐次通解 $\vec{x} = C_1 \vec{r}_1 + C_2 \vec{r}_2 + \vec{x}^* = \begin{bmatrix} C_1 e^t + 2C_2 e^{2t} + 23e^{2t} \\ 2C_1 e^t + 3C_2 e^{2t} + 3e^{2t} \end{bmatrix}$

Summary 常系数线性方程组 20%

$$\begin{cases} \frac{dx}{dt} = 5x - y + e^{2t} \\ \frac{dy}{dt} = 6x - 2y \end{cases} \quad \begin{cases} \frac{dx}{dt} = -y \\ \frac{dy}{dt} = x \end{cases}$$

2未知函数 消元法

求齐次/非齐次方程组

由一未知 3未知函数 特征向量法

求齐次方程组通解

常数变易法

求非齐次方程组特解

幂级数法

$$dy/dx = y$$

$$\text{令 } y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots \Rightarrow \begin{cases} a_1 = a_0 \\ a_2 = 2a_1 \\ a_3 = 3a_2 \\ \dots \end{cases} \Rightarrow \begin{cases} a_1 = a_0 \\ a_2 = a_0/2 \\ a_3 = a_0/6 \\ \dots \end{cases} \Rightarrow y = a_0 \left(1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \dots \right) \Rightarrow y = Ce^x$$

① 一阶方程 $\frac{dy}{dx} = f(x, y)$ 30-40%

变量分离 $\frac{dy}{dx} = \varphi(x)\phi(y)$ $e^x y' = \sqrt{y} \ln y$

变量型齐次 $\frac{dy}{dx} = g(\frac{y}{x})$ $\frac{dy}{dx} = \frac{y}{x+y}$

一阶线性 $\frac{dy}{dx} + P(x)y = f(x)$ $y' + \frac{y}{x} = 2$ $(x+y)dy = dx$

全微分 $Mdx + Ndy = 0$ $x dx + y dy = 0$ $x dy - y dx = 0$

② 二阶方程 $y'' = f(x, y, y')$ 30-40%

可降阶 $y'' = \cos x$ $y'' = 2x(y')^2$ $y'' = 2y(y')^3$

二阶常系数线性 $y'' + 4y' - 5y = 0$ $y'' + 4y' - 5y = 16e^{3x}$
 $y'' + 4y' + 4y = 0$ $y'' + 4y' + 4y = 6xe^{2x}$
 $y'' + 4y' + 7y = 0$ $y'' - y = e^x \sin 2x$

二阶变系数线性 $x^2 y''' + x y' + 4y = x^2$ Euler
 $y'' + \frac{1}{x} y' - \frac{1}{x^2} y = 0$ Liouville
 $y'' - 3y' + 2y = \frac{1}{1+e^x}$ 常数变易

变形及向高阶的推广 P_{34-2} P_{30-15} $P_{120-121}$

③ 常系数线性方程组 20%

$\begin{cases} \frac{dx}{dt} = 5x - 2y + e^{2t} \\ \frac{dy}{dt} = 6x - 2y \end{cases}$ $\begin{cases} \frac{dx}{dt} = -y \\ \frac{dy}{dt} = x \end{cases}$

2未知函数 消元法

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